Example: Toss a coin thrice

A fair coin is tossed thrice. Naturally, there can be three random variables.

Let $X_i = 1$ if i-th toss is heads and $X_i = 0$ if i-th toss is tails, $i = 1, 2, 3$.

Example: Toss a coin thrice

A fair coin is tossed thrice. Naturally, there can be three random variables.

Let $X_i = 1$ if i-th toss is heads and $X_i = 0$ if i-th toss is tails, $i = 1, 2, 3$.

Together, the 3 random variables completely describe the outcome of the experiment.

The event $X_1 = 1$ is independent of $X_2 = 1$ and $X_3 = 1$.

Example: Random 2-digit number

A 2-digit number from 00 to 99 is selected at random. Partial information is available about the number as two random variables. Let X be the digit in units place. Let Y be the reminder obtained when the number is divided by 4.

$23 \Rightarrow X=3$
$Y=3$
$48 \Rightarrow X=8$
$Y=0$

$$X \in \text{Uniform}(\{0,1,2,3,4,5,6,7,8,9\})$$
$$Y \in \text{Uniform}(\{0,1,2,3\})$$

Example: Random 2-digit number

A 2-digit number from 00 to 99 is selected at random. Partial information is available about the number as two random variables. Let X be the digit in units place. Let Y be the reminder obtained when the number is divided by 4.

$$X \in \text{Uniform}(\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\})$$
$$Y \in \text{Uniform}(\{0, 1, 2, 3\})$$

Suppose the event $X = 1$ has occurred. What about the event $Y = 0$?

Example: Random 2-digit number

A 2-digit number from 00 to 99 is selected at random. Partial information is available about the number as two random variables. Let X be the digit in units place. Let Y be the reminder obtained when the number is divided by 4.

$$X \in \text{Uniform}(\{0,1,2,3,4,5,6,7,8,9\})$$
$$Y \in \text{Uniform}(\{0,1,2,3\})$$

Suppose the event $X = 1$ has occurred. What about the event $Y = 0$?

When two random variables are defined in the same probability space, the value of one can influence the value of the other.

Example: IPL powerplay over

Let $X =$ number of runs in the over. Let $Y =$ number of wickets in the over.

Consider the events: $Y = 0$, $Y = 1$, $Y = 2$

Example: IPL powerplay over

Let X = number of runs in the over. Let Y = number of wickets in the over.

Consider the events: $Y = 0$, $Y = 1$, $Y = 2$

Given $Y = 0$, we expect X to take larger values than when $Y = 1$.

Given $Y = 2$, we expect X to take significantly lower values.

Example: IPL powerplay over

Let $X =$ number of runs in the over. Let $Y =$ number of wickets in the over.

Consider the events: $Y = 0$, $Y = 1$, $Y = 2$

Given $Y = 0$, we expect X to take larger values than when $Y = 1$.

Given $Y = 2$, we expect X to take significantly lower values.

In complex experiments, such relationships between random variables are useful in modeling.

Section 1

Two random variables: Joint, marginal, conditional PMFs

Two discrete random variables: Joint PMF

Definition (Joint PMF)

Suppose X and Y are discrete random variables defined in the same probability space. Let the range of X and Y be T_X and T_Y, respectively. The joint PMF of X and Y, denoted f_{XY}, is a function from $T_X \times T_Y$ to $[0, 1]$ defined as

$$f_{XY}(t_1, t_2) = P(X = t_1 \text{ and } Y = t_2), t_1 \in T_X, t_2 \in T_Y.$$

- Joint PMF is usually written as a table or a matrix
- $P(X = t_1 \text{ and } Y = t_2)$ is denoted $\underline{P(X = t_1, Y = t_2)}$

Example: Toss a fair coin twice

Let $X_i = 1$ if i-th toss is heads and $X_i = 0$ if i-th toss is tails, $i = 1, 2$.

- $f_{X_1 X_2}(0, 0) = P(X_1 = 0 \text{ and } X_2 = 0) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$
- $f_{X_1 X_2}(0, 1) = P(X_1 = 0 \text{ and } X_2 = 1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

Example: Toss a fair coin twice

Let $X_i = 1$ if i-th toss is heads and $X_i = 0$ if i-th toss is tails, $i = 1, 2$.

- $f_{X_1 X_2}(0,0) = P(X_1 = 0 \text{ and } X_2 = 0) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$
- $f_{X_1 X_2}(0,1) = P(X_1 = 0 \text{ and } X_2 = 1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

Table 1: $f_{X_1 X_2}(t_1, t_2)$

$t_2 \setminus t_1$	0	1
0	1/4	1/4
1	1/4	1/4 → $f_{X_1X_2}(1,1)$

Joint PMF ⇒ { - Each entry is between 0 + 1
- Sum of all entries equals 1 }

Example: Random 2-digit number

X = units place, Y = number modulo 4 remainder when divided by 4

- $f_{XY}(0,0) = P(X = 0 \text{ and } Y = 0)$

  $= P(\text{number ends in 0 and multiple of 4})$

  $= P(\{00, 20, 40, 60, 80\}) = 5/100 = 1/20$

- $f_{XY}(1,0) = P(\text{number ends in 1 and multiple of 4}) = 0$

  $X=1$ $Y=0$

- $f_{XY}(4,2) = P(\text{number ends in 4 and 2 mod 4})$

  $X=4$ $Y=2$

  $= P(\{14, 34, 54, 74, 94\}) = 5/100 = 1/20$

Example: Random 2-digit number

X = units place, Y = number modulo 4

- $f_{XY}(0,0) = P(X = 0 \text{ and } Y = 0)$

  $= P(\text{number ends in 0 and multiple of 4})$

  $= P(\{00, 20, 40, 60, 80\}) = 5/100 = 1/20$

- $f_{XY}(1,0) = P(\text{number ends in 1 and multiple of 4}) = 0$

- $f_{XY}(4,2) = P(\text{number ends in 2 and 2 mod 4})$
  $= P(\{14, 34, 54, 74, 94\}) = 5/100 = 1/20$

Table of $f_{XY}(t_1, t_2)$

$t_2 \setminus t_1$	0	1	2	3	4	5	6	7	8	9
0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0
1	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$
2	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0
3	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$

Two random variables: Marginal PMF

Definition (Marginal PMF)

Suppose X and Y are jointly distributed discrete random variables with joint PMF f_{XY}. The PMF of the individual random variables X and Y are called as marginal PMFs. It can be shown that

$$f_X(t) = P(X = t) = \sum_{t' \in T_Y} f_{XY}(t, t'),$$

Marginalisation

$$f_Y(t) = P(Y = t) = \sum_{t' \in T_X} f_{XY}(t', t),$$

where T_X and T_Y are the ranges of X and Y, repectively.

- Proof
  - Suppose $T_Y = \{y_1, \ldots, y_K\}$
  - $(\underline{X = t}) = (X = t \text{ and } Y = y_1) \text{ or } \cdots \text{ or } (X = t \text{ and } Y = y_K)$
  - $P(X = t) = P(X = t, Y = y_1) + \cdots + P(X = t, Y = y_K)$
- Note that the marginal PMF is simply a PMF

Example: Toss a fair coin twice

Table of $f_{X_1 X_2}(t_1, t_2)$

$t_2 \setminus t_1$	0	1	$f_{X_2}(t_2)$
0	1/4	1/4	1/2
1	1/4	1/4	1/2
$f_{X_1}(t_1)$	1/2	1/2	

- Marginal PMF of X_1: add over the columns of joint PMF table
  - $f_{X_1}(0) = f_{X_1 X_2}(0, 0) + f_{X_1 X_2}(0, 1)$
  - $f_{X_1}(1) = f_{X_1 X_2}(1, 0) + f_{X_1 X_2}(1, 1)$
- Marginal PMF of X_2: add over the rows of joint PMF table
  - $f_{X_2}(0) = f_{X_1 X_2}(0, 0) + f_{X_1 X_2}(1, 0)$
  - $f_{X_2}(1) = f_{X_1 X_2}(0, 1) + f_{X_1 X_2}(1, 1)$

Example: Marginal from joint PMF

Table of $f_{X_1 X_2}(t_1, t_2)$

$t_2 \setminus t_1$	0	1	$f_{X_2}(t_2)$
0	0.05	0.35	0.40
1	0.25	0.35	0.60
$f_{X_1}(t_1)$	0.30	0.70	

Verify: valid Joint PMF

$0.05 + 0.35 + 0.25 + 0.35 = 1$ ✓

Example: Same marginal PMF from different joint PMFs

Case 1

$t_2 \setminus t_1$	0	1	$f_{X_2}(t_2)$
0	$1/4$	$1/4$	$1/2$
1	$1/4$	$1/4$	$1/2$
$f_{X_1}(t_1)$	$1/2$	$1/2$	

Case 2

$t_2 \setminus t_1$	0	1	$f_{X_2}(t_2)$
0	x	$1/2 - x$	$1/2$
1	$1/2 - x$	x	$1/2$
$f_{X_1}(t_1)$	$1/2$	$1/2$	

- For every x between 0 and $1/2$, we get a joint PMF that results in the same marginal.

Example: Random 2-digit number

Table of $f_{XY}(t_1, t_2)$

$t_2 \setminus t_1$	0	1	2	3	4	5	6	7	8	9	$f_Y(t_2)$
0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$1/4$
1	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	$1/4$
2	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$1/4$
3	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	0	$\frac{1}{20}$	$1/4$
$f_X(t_1)$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	

$X \sim \text{Uniform}\{0, 1, \ldots, 9\}$

$Y \sim \text{Uniform}\{0, 1, 2, 3\}$

$P(X=0, Y=0) = \Pr(X=0 \text{ and } Y=0) = 1/20$

$P(X=0)\,P(Y=0) = \frac{1}{10} \cdot \frac{1}{4} = \frac{1}{40}$

Not equal

Conditional distribution of a random variable given an event

Definition (Conditional distribution given an event)

Suppose X is a discrete random variable with range T_X, and A is an event in the same probability space. The conditional PMF of X given A is defined as the PMF

$$Q(t) = P(X = t|A), \qquad t \in T_X.$$

We will use the notation $f_{X|A}(t)$ for the above conditional PMF, and $(X|A)$ to denote the "conditional" random variable.

Conditional distribution of a random variable given an event

Definition (Conditional distribution given an event)

Suppose X is a discrete random variable with range T_X, and A is an event in the same probability space. The conditional PMF of X given A is defined as the PMF

$$Q(t) = P(X = t|A), \qquad t \in T_X.$$

We will use the notation $f_{X|A}(t)$ for the above conditional PMF, and $(X|A)$ to denote the "conditional" random variable.

$$f_{X|A}(t) = \frac{P((X = t) \cap A)}{P(A)}$$

- Important: Range of $(X|A)$ can be different from T_X and will depend on A

Conditional distribution of one random variable given another

Definition (Conditional distribution of Y given $X = t$)

Suppose X and Y are jointly distributed discrete random variables with joint PMF f_{XY}. The conditional PMF of Y given $X = t$ is defined as the PMF

$$Q(t') = P(Y = t'|X = t) = \frac{P(Y = t', X = t)}{P(X = t)} = \frac{f_{XY}(t, t')}{f_X(t)}.$$

We will use the notation $f_{Y|X=t}(t')$ for the above conditional PMF, and $(Y|X = t)$ to denote the "conditional" random variable.

Conditional distribution of one random variable given another

Definition (Conditional distribution of Y given $X = t$)

Suppose X and Y are jointly distributed discrete random variables with joint PMF f_{XY}. The conditional PMF of Y given $X = t$ is defined as the PMF

$$Q(t') = P(Y = t'|X = t) = \frac{P(Y = t', X = t)}{P(X = t)} = \frac{f_{XY}(t, t')}{f_X(t)}.$$

We will use the notation $f_{Y|X=t}(t')$ for the above conditional PMF, and $(Y|X = t)$ to denote the "conditional" random variable.

$$\boxed{f_{XY}(t, t') = f_{Y|X=t}(t') f_X(t)}$$

- Important: Range of $(Y|X = t)$ can be different from range of Y and will depend on t

Example: Compute conditional PMFs from joint PMF

Joint PMF $f_{XY}(t_1, t_2)$

$t_2 \setminus t_1$	0	1	2	$f_Y(t_2)$
0	1/4	1/8	1/8	1/2
1	1/8	1/8	1/4	1/2
$f_X(t_1)$	3/8	1/4	3/8	

$X \in \{0, 1, 2\}$, $Y \in \{0, 1\}$

$(Y|X=0) \in \{0, 1\}$

$f_{Y|X=0}(0) = \dfrac{f_{XY}(0,0)}{f_X(0)} = \dfrac{1/4}{3/8} = 2/3$

$f_{Y|X=0}(1) = \dfrac{f_{XY}(0,1)}{f_X(0)} = \dfrac{1/8}{3/8} = 1/3$

$(X|Y=1) \in \{\overset{\frac{1/8}{1/2} = \frac{1}{4}}{0}, \overset{\frac{1/8}{1/2} = \frac{1}{4}}{1}, \overset{\frac{1/4}{1/2} = \frac{1}{2}}{2}\}$

Example: Compute marginal/conditional PMFs from joint PMF

Joint PMF $f_{XY}(t_1, t_2)$

$t_2 \setminus t_1$	0	1	2	$f_Y(t_2)$
0	1/12	0	3/12	$1/3$
1	2/12	1/12	0	$1/4$
2	3/12	1/12	1/12	$5/12$
$f_X(t_1)$	$1/2$	$1/6$	$1/3$	

$$\sum_{t' \in T_Y} f_{Y|X=t}(t') = 1$$

$$Y|X=0 \sim \left\{ \overset{1/6}{0}, \overset{1/3}{1}, \overset{1/2}{2} \right\}$$

$$Y|X=1 \sim \left\{ \overset{1/2}{1}, \overset{1/2}{2} \right\}, \quad Y|X=2 \sim \left\{ \overset{3/4}{0}, \overset{1/4}{2} \right\}$$

$$X|Y=1 \sim \left\{ \overset{2/3}{0}, \overset{1/3}{1} \right\}$$

$$\boxed{f_{XY}(t_1, t_2) = f_{Y|X=t_1}(t_2)\, f_X(t_1) = f_{X|Y=t_2}(t_1) \cdot f_Y(t_2)}$$

Example: Throw a die and toss coins

Throw a die and toss a coin as many times as the number shown on die. Let X be the number shown on die. Let Y be the number of heads. What is the joint PMF of X and Y?

$$X \sim \text{Uniform}(1,2,3,4,5,6) \qquad f_X(t) = \frac{1}{6} \quad 1 \le t \le 6$$

$$(Y|X=t) \sim \text{Binomial}(t, 1/2) \qquad f_{Y|X=t}(t') = {}^tC_{t'} \left(\tfrac{1}{2}\right)^{t'} \cdot \left(\tfrac{1}{2}\right)^{t-t'} = {}^tC_{t'} \left(\tfrac{1}{2}\right)^{t} \quad t' = 0, 1, 2, \ldots, t$$

$$f_{XY}(t,t') = f_X(t)\, f_{Y|X=t}(t') = \frac{1}{6} \cdot {}^tC_{t'} \left(\tfrac{1}{2}\right)^{t}$$

$$\text{Range}(Y|X=t) = \{0, 1, \ldots, t\}$$

$t' \backslash t$	1	2	3	4	5	6
0	$\frac{1}{6}\cdot \frac{1}{2}$	$\frac{1}{6}\cdot\left(\frac{1}{2}\right)^2$				
1	$\frac{1}{6}\cdot \frac{1}{2}$	$\frac{1}{6}\cdot 2\left(\frac{1}{2}\right)^2$				
2	0	$\frac{1}{6}\left(\frac{1}{2}\right)^2$				
3	0	0				
⋮	⋮	⋮				

Exercise

Example: Poisson number of coin tosses

Let $N \sim \text{Poisson}(\lambda)$. Given $N = n$, toss a fair coin n times and denote the number of heads obtained by X. What is the distribution of X?

$f_X(x) = ?$

$$f_N(n) = \frac{e^{-\lambda} \lambda^n}{n!}, \quad n = 0, 1, 2, \ldots$$

$$(X \mid N = n) \sim \text{Binomial}(n, 1/2) \qquad f_{X|N=n}(k) = \binom{n}{k}\left(\frac{1}{2}\right)^n, \quad k = 0, 1, \ldots, n$$

$$f_{NX}(n, k) = \frac{e^{-\lambda} \lambda^n}{n!} \cdot \binom{n}{k}\left(\frac{1}{2}\right)^n \qquad \begin{array}{l} n = 0, 1, 2, \ldots \\ k = 0, 1, \ldots, n \end{array}$$

$${}^nC_k = \binom{n}{k}$$

$$\underbrace{f_X(k)}_{\text{Got } k \text{ heads}} = \sum_{n=0}^{\infty} \underbrace{f_{NX}(n, k)}_{=0 \text{ if } n < k} = \sum_{n=k}^{\infty} \frac{e^{-\lambda} \lambda^n}{n!} \binom{n}{k}\left(\frac{1}{2}\right)^n$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$$= \sum_{n=k}^{\infty} \frac{e^{-\lambda} \lambda^n}{k!(n-k)!} \left(\frac{1}{2}\right)^n$$

$$= \frac{e^{-\lambda} \lambda^k}{k! \, 2^k} \underbrace{\sum_{n=k}^{\infty} \frac{\lambda^{n-k}}{(n-k)! \cdot 2^{n-k}}}_{e^{\lambda/2}} \qquad (\lambda/2)^{n-k}$$

$$\boxed{X \sim \text{Poisson}(\lambda/2)}$$

$$\boxed{f_X(k) = \frac{e^{-\lambda/2} (\lambda/2)^k}{k!}} \quad k = 0, 1, 2, \ldots$$

Example: IPL powerplay over

Let X = number of runs in the over. Let Y = number of wickets in the over. Assume the following:

$$Y \sim \left\{ \overset{13/16}{0}, \overset{1/8}{1}, \overset{1/16}{2} \right\},$$

$$(X|Y=0) \sim \text{Uniform}\{6, 7, 8, 9, 10, 11, 12\},$$

$$(X|Y=1) \sim \text{Uniform}\{2, 3, 4, 5, 6, 7, 8\},$$

$$(X|Y=2) \sim \text{Uniform}\{0, 1, 2, 3, 4, 5, 6\}.$$

$f_{XY}(t,t') = f_Y(t') \cdot f_{X|Y=t'}(t)$

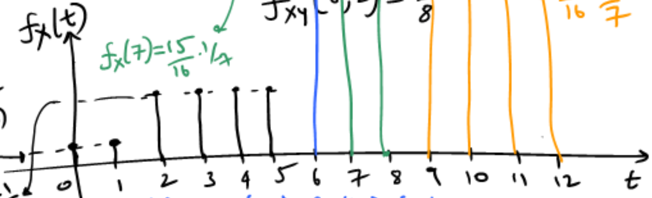
$f_{XY}(6,1) = \frac{1}{8}\cdot\frac{1}{7}$

$f_{XY}(8,2) = \frac{1}{8}$

$f_X(t) = ?$

$t = 0, 1, 2, \dots, 12$

$f_X(7) = \frac{15}{16}\cdot 1/7$



$f_X(0) = f_{XY}(0,2) = \frac{1}{16}\cdot\frac{1}{7} = f_X(1)$

$f_X(2) = \underbrace{f_{XY}(2,2)}_{\frac{1}{16}\cdot\frac{1}{7}} + \underbrace{f_{XY}(2,1)}_{\frac{1}{8}\cdot\frac{1}{7}} = \frac{3}{16}\cdot\frac{1}{7}$

$f_X(6) = f_{XY}(6,2) + f_{XY}(6,1) + f_{XY}(6,0) = 1/7$

Section 2

More than two random variables: Joint, marginal, conditional PMFs

Multiple discrete random variables: Joint PMF

Definition (Joint PMF)

Suppose $X_1, X_2, \ldots, X_n$ are discrete random variables defined in the same probability space. Let the range of X_i be T_{X_i}. The joint PMF of X_i, denoted $f_{X_1 \cdots X_n}$, is a function from $T_{X_1} \times \cdots \times T_{X_n}$ to $[0, 1]$ defined as

$$f_{X_1 \cdots X_n}(t_1, \ldots, t_n) = P(X_1 = t_1 \text{ and } \cdots \text{ and } X_n = t_n), t_i \in T_{X_i}.$$

- Joint PMF could be written as a table in small examples
- $P(X_1 = t_1 \text{ and } \cdots \text{ and } X_n = t_n)$ is denoted $P(X_1 = t_1, \ldots, X_n = t_n)$

Example: Toss a fair coin thrice

Let $X_i = 1$ if i-th toss is heads and $X_i = 0$ if i-th toss is tails, $i = 1, 2, 3$.

t_1	t_2	t_3	$f_{X_1X_2X_3}(t_1, t_2, t_3)$
0	0	0	1/8
0	0	1	1/8
0	1	0	1/8
0	1	1	1/8
1	0	0	1/8
1	0	1	1/8
1	1	0	1/8
1	1	1	1/8

Example: Random 3-digit number 000 to 999

$X =$ first digit from left, $Y =$ number modulo 2, $Z =$ first digit from right

- Table is too long

$$X \in \{0, 1, 2, \ldots, 9\}$$

$$Y \in \{0, 1\}$$

$$Z \in \{0, 1, 2, \ldots, 9\}$$

$f_{XYZ}(0,0,0) = P($Starts with zero and number is even and number ends in zero$)$

$$= \frac{10}{1000} = \frac{1}{100}$$

$f_{XYZ}(1,1,1) = \frac{1}{100}$, $f_{XYZ}(1,0,1) = 0$

$f_{XYZ}(8,0,6) = \frac{10}{1000} = \frac{1}{100}$, $f_{XYZ}(8,1,6) = 0$

$$f_{XYZ}(t_1,t_2,t_3) = \begin{cases} 0 & \text{if } t_2 = 0 \text{ \& } t_3\text{: odd} \\ 0 & \text{if } t_2 = 1 \text{ \& } t_3\text{: even} \\ \frac{1}{100} & \text{otherwise} \end{cases}$$

Example: IPL powerplay over

Suppose the over has 6 deliveries. Let X_i denote the number of runs scored in the i-th delivery.

$$X_i \in \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$$

$i = 1, 2, \ldots, 6$

Joint PMF

$$\underbrace{f_{X_1X_2X_3X_4X_5X_6}}_{8^6 \text{: too many possibilities} \ldots\ldots}(0, 1, 4, 2, 0, 0) = ?$$

Multiple discrete random variables: Marginal PMF (individual)

Definition (Marginal PMF (individual))

Suppose $X_1, X_2, \ldots, X_n$ are jointly distributed discrete random variables with joint PMF $f_{X_1 \cdots X_n}$. The PMF of the individual random variables X_1, X_2, $\cdots$, X_n are called as marginal PMFs. It can be shown that

$$f_{X_1}(t) = P(X_1 = t) = \sum_{t_2' \in T_{X_2}, t_3' \in T_{X_3}, \ldots, t_n' \in T_{X_n}} f_{X_1 \cdots X_n}(t, t_2', t_3', \ldots, t_n'),$$

$$f_{X_2}(t) = P(X_2 = t) = \sum_{t_1' \in T_{X_1}, t_3' \in T_{X_3}, \ldots, t_n' \in T_{X_n}} f_{X_1 \cdots X_n}(t_1', t, t_3', \ldots, t_n'),$$

$$\vdots$$

$$f_{X_n}(t) = P(X_n = t) = \sum_{t_1' \in T_{X_1}, \ldots, t_{n-1}' \in T_{X_{n-1}}, t \in T_{X_n}} f_{X_1 \cdots X_n}(t_1', \ldots, t_{n-1}', t),$$

where T_{X_i} is the range of X_i.

Example: Toss a fair coin thrice

Let $X_i = 1$ if i-th toss is heads and $X_i = 0$ if i-th toss is tails, $i = 1, 2, 3$.

Joint PMF: $f_{X_1 X_2 X_3}(t_1, t_2, t_3) = 1/8$ for $t_i \in \{0, 1\}$

Example: Toss a fair coin thrice

Let $X_i = 1$ if i-th toss is heads and $X_i = 0$ if i-th toss is tails, $i = 1, 2, 3$.

Joint PMF: $f_{X_1X_2X_3}(t_1, t_2, t_3) = 1/8$ for $t_i \in \{0, 1\}$

$$\begin{aligned} f_{X_1}(0) &= f_{X_1X_2X_3}(0, 0, 0) + f_{X_1X_2X_3}(0, 0, 1) + f_{X_1X_2X_3}(0, 1, 0) + f_{X_1X_2X_3}(0, 1, 1) \\ &= 1/8 + 1/8 + 1/8 + 1/8 = 1/2 \\ f_{X_1}(1) &= f_{X_1X_2X_3}(1, 0, 0) + f_{X_1X_2X_3}(1, 0, 1) + f_{X_1X_2X_3}(1, 1, 0) + f_{X_1X_2X_3}(1, 1, 1) \\ &= 1/8 + 1/8 + 1/8 + 1/8 = 1/2 \end{aligned}$$

Note: $f_{X_1}(0) + f_{X_1}(1) = 1$

Example: Random 3-digit number 000 to 999

X = first digit from left, Y = number modulo 2, Z = first digit from right

- $X \sim \text{Uniform}\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

Example: Random 3-digit number 000 to 999

$X =$ first digit from left, $Y =$ number modulo 2, $Z =$ first digit from right

- $X \sim \text{Uniform}\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- $Y \sim \text{Uniform}\{0, 1\}$

Example: Random 3-digit number 000 to 999

$X =$ first digit from left, $Y =$ number modulo 2, $Z =$ first digit from right

- $X \sim \text{Uniform}\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- $Y \sim \text{Uniform}\{0, 1\}$
- $Z \sim \text{Uniform}\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

Example: IPL powerplay Over 1

Suppose the over has 6 deliveries. Let X_i denote the number of runs scored in the i-th delivery.

Example: IPL powerplay Over 1

Suppose the over has 6 deliveries. Let X_i denote the number of runs scored in the i-th delivery.

$X_i \in \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$: How to assign probabilities?

Example: IPL powerplay Over 1

Suppose the over has 6 deliveries. Let X_i denote the number of runs scored in the i-th delivery.

$X_i \in \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$: How to assign probabilities?

Ball 1: 0 - 957 matches, 1 - 429 matches, 2 - 57 matches, 3 - 5 matches, 4 - 138 matches, 5 - 8 matches, 6 - 4 matches (out of 1598 matches)

Idea: Assign probabilities in the same proportion as data

Example: IPL powerplay Over 1

Suppose the over has 6 deliveries. Let X_i denote the number of runs scored in the i-th delivery.

$X_i \in \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$: How to assign probabilities?

Ball 1: 0 - 957 matches, 1 - 429 matches, 2 - 57 matches, 3 - 5 matches, 4 - 138 matches, 5 - 8 matches, 6 - 4 matches (out of 1598 matches)

Idea: Assign probabilities in the same proportion as data

	0	1	2	3	4	5	6
f_{X_1}	0.5989	0.2685	0.0357	0.0031	0.0864	0.0050	0.0025
f_{X_2}	0.5551	0.2791	0.0438	0.0031	0.1083	0.0044	0.0063
f_{X_3}	0.5338	0.2847	0.0444	0.0044	0.1139	0.0025	0.0163
f_{X_4}	0.5344	0.2516	0.0394	0.0031	0.1489	0.0038	0.0188
f_{X_5}	0.5313	0.2672	0.0407	0.0056	0.1358	0.0025	0.0169
f_{X_6}	0.5056	0.2954	0.0394	0.0050	0.1414	0.0013	0.0119

Marginalisation

Suppose $X_1, X_2, X_3 \sim f_{X_1X_2X_3}$ and $X_i \in T_{X_i}$.

We have discussed the marginal PMF f_{X_i} of the individual random variables X_1, X_2 and X_3.

What about $f_{X_1X_2}$, the joint PMF of X_1 and X_2? What about $f_{X_1X_3}$, $f_{X_2X_3}$?

Marginalisation

Suppose $X_1, X_2, X_3 \sim f_{X_1X_2X_3}$ and $X_i \in T_{X_i}$.

We have discussed the marginal PMF f_{X_i} of the individual random variables X_1, X_2 and X_3.

What about $f_{X_1X_2}$, the joint PMF of X_1 and X_2? What about $f_{X_1X_3}$, $f_{X_2X_3}$?

$$f_{X_1X_2}(t_1, t_2) = P(X_1 = t_1 \text{ and } X_2 = t_2) = \sum_{t_3' \in T_{X_3}} f_{X_1X_2X_3}(t_1, t_2, t_3')$$

$$f_{X_1X_3}(t_1, t_3) = P(X_1 = t_1 \text{ and } X_3 = t_3) = \sum_{t_2' \in T_{X_2}} f_{X_1X_2X_3}(t_1, t_2', t_3)$$

$$f_{X_2X_3}(t_2, t_3) = P(X_2 = t_2 \text{ and } X_3 = t_3) = \sum_{t_1' \in T_{X_1}} f_{X_1X_2X_3}(t_1', t_2, t_3)$$

Marginalisation

Suppose $X_1, X_2, X_3 \sim f_{X_1X_2X_3}$ and $X_i \in T_{X_i}$.

We have discussed the marginal PMF f_{X_i} of the individual random variables X_1, X_2 and X_3.

What about $f_{X_1X_2}$, the joint PMF of X_1 and X_2? What about $f_{X_1X_3}$, $f_{X_2X_3}$?

$$f_{X_1X_2}(t_1, t_2) = P(X_1 = t_1 \text{ and } X_2 = t_2) = \sum_{t_3' \in T_{X_3}} f_{X_1X_2X_3}(t_1, t_2, t_3')$$

$$f_{X_1X_3}(t_1, t_3) = P(X_1 = t_1 \text{ and } X_3 = t_3) = \sum_{t_2' \in T_{X_2}} f_{X_1X_2X_3}(t_1, t_2', t_3)$$

$$f_{X_2X_3}(t_2, t_3) = P(X_2 = t_2 \text{ and } X_3 = t_3) = \sum_{t_1' \in T_{X_1}} f_{X_1X_2X_3}(t_1', t_2, t_3)$$

Marginalisation: Sum over everything you do not want!

Example: $X_1, X_2, X_3 \sim f_{X_1X_2X_3}$

t_1	t_2	t_3	$f_{X_1X_2X_3}(t_1, t_2, t_3)$
0	0	0	1/9
0	0	1 ✓	1/9
0	0	2 *	1/9
0	1	1 ✓	1/9
0	1	2 *	1/9
1	0	0 *	1/9
1	0	2	1/9
1	1	0 *	1/9
1	1	1	1/9

$f_{X_1X_2}$

$t_2 \backslash t_1$	0	1
0	1/3	2/9
1	2/9	2/9

$f_{X_1X_3}$

$t_2 \backslash t_1$	0	1
0	1/9	2/9 *
1	2/9 ✓	1/9
2	2/9 *	1/9

Working

Marginalisation: More examples

Suppose $X_1, X_2, X_3, X_4 \sim f_{X_1X_2X_3X_4}$ and $X_i \in T_{X_i}$.

$$f_{X_1}(t_1) = P(X_1 = t_1) = \sum_{t_2', t_3', t_4'} f_{X_1X_2X_3X_4}(t_1, t_2', t_3', t_4')$$

$$f_{X_1X_2}(t_1, t_2) = P(X_1 = t_1 \text{ and } X_2 = t_2) = \sum_{t_3', t_4'} f_{X_1X_2X_3X_4}(t_1, t_2, t_3', t_4')$$

$$f_{X_2X_4}(t_2, t_4) = P(X_2 = t_2 \text{ and } X_4 = t_4) = \sum_{t_1', t_3'} f_{X_1X_2X_3X_4}(t_1', t_2, t_3', t_4)$$

$$f_{X_1X_3X_4}(t_1, t_3, t_4) = \sum_{t_2'} f_{X_1X_2X_3X_4}(t_1, t_2', t_3.t_4)$$

Marginalisation: Sum over everything you do not want!

Multiple discrete random variables: Marginal PMF (general)

Definition (Marginal PMF (general))

Suppose $X_1, X_2, \ldots, X_n$ are jointly distributed discrete random variables with joint PMF $f_{X_1 \cdots X_n}$. The joint PMF of the random variables $X_{i_1}, X_{i_2}, \ldots, X_{i_k}$, denoted $f_{X_{i_1} \cdots X_{i_k}}$, is given by

$$f_{X_{i_1} \cdots X_{i_k}}(t_{i_1}, \ldots, t_{i_k}) = \sum_{\substack{t_1, \ldots, t'_{i_1-1}, t'_{i_1+1}, \ldots, \\ \ldots, t'_{i_k-1}, t'_{i_k+1}, \ldots, t_n}} f_{X_1 \cdots X_n}(t_1, \ldots, t'_{i_1-1}, t_{i_1}, t'_{i_1+1}, \ldots, t'_{i_k-1}, t_{i_k}, t'_{i_k+1}, \ldots, t_n).$$

Conditioning with multiple discrete random variables

Suppose $X_1, X_2, X_3, X_4 \sim f_{X_1X_2X_3X_4}$ and $X_i \in T_{X_i}$.

A wide variety of conditioning is possible when there are many random variables.

Conditioning with multiple discrete random variables

Suppose $X_1, X_2, X_3, X_4 \sim f_{X_1X_2X_3X_4}$ and $X_i \in T_{X_i}$.

A wide variety of conditioning is possible when there are many random variables.

$$(X_1|X_2 = t_2) \sim f_{X_1|X_2=t_2}(t_1) = \frac{f_{X_1X_2}(t_1, t_2)}{f_{X_2}(t_2)}$$

Conditioning with multiple discrete random variables

Suppose $X_1, X_2, X_3, X_4 \sim f_{X_1X_2X_3X_4}$ and $X_i \in T_{X_i}$.

A wide variety of conditioning is possible when there are many random variables.

$$(X_1|X_2 = t_2) \sim f_{X_1|X_2=t_2}(t_1) = \frac{f_{X_1X_2}(t_1, t_2)}{f_{X_2}(t_2)}$$
$$(X_1, X_2|X_3 = t_3) \sim f_{X_1X_2|X_3=t_3}(t_1, t_2) = \frac{f_{X_1X_2X_3}(t_1, t_2, t_3)}{f_{X_3}(t_3)}$$

Conditioning with multiple discrete random variables

Suppose $X_1, X_2, X_3, X_4 \sim f_{X_1X_2X_3X_4}$ and $X_i \in T_{X_i}$.

A wide variety of conditioning is possible when there are many random variables.

$$(X_1|X_2 = t_2) \sim f_{X_1|X_2=t_2}(t_1) = \frac{f_{X_1X_2}(t_1, t_2)}{f_{X_2}(t_2)}$$

$$(X_1, X_2|X_3 = t_3) \sim f_{X_1X_2|X_3=t_3}(t_1, t_2) = \frac{f_{X_1X_2X_3}(t_1, t_2, t_3)}{f_{X_3}(t_3)}$$

$$(X_1|X_2 = t_2, X_3 = t_3) \sim f_{X_1|X_2=t_2,X_3=t_3}(t_1) = \frac{f_{X_1X_2X_3}(t_1, t_2, t_3)}{f_{X_2X_3}(t_2, t_3)}$$

Conditioning with multiple discrete random variables

Suppose $X_1, X_2, X_3, X_4 \sim f_{X_1X_2X_3X_4}$ and $X_i \in T_{X_i}$.

A wide variety of conditioning is possible when there are many random variables.

$$(X_1|X_2 = t_2) \sim f_{X_1|X_2=t_2}(t_1) = \frac{f_{X_1X_2}(t_1, t_2)}{f_{X_2}(t_2)}$$

$$(X_1, X_2|X_3 = t_3) \sim f_{X_1X_2|X_3=t_3}(t_1, t_2) = \frac{f_{X_1X_2X_3}(t_1, t_2, t_3)}{f_{X_3}(t_3)}$$

$$(X_1|X_2 = t_2, X_3 = t_3) \sim f_{X_1|X_2=t_2,X_3=t_3}(t_1) = \frac{f_{X_1X_2X_3}(t_1, t_2, t_3)}{f_{X_2X_3}(t_2, t_3)}$$

$$(X_1, X_3|X_2 = t_2, X_4 = t_4) \sim f_{X_1X_3|X_2=t_2,X_4=t_4}(t_1, t_3) = \frac{f_{X_1X_2X_3X_4}(t_1, t_2, t_3, t_4)}{f_{X_2X_4}(t_2, t_4)}$$

Example: $X_1, X_2, X_3, X_4 \sim f_{X_1X_2X_3X_4}$

t_1	t_2	t_3	t_4	$f_{X_1 \cdots X_4}(t_1, \ldots, t_4)$
0	0	0	0	1/12
0	0	0	1	1/12
0	0	1	1	1/12
0	0	2	0	1/12
0	1	1	0	1/12
0	1	1	1	1/12
0	1	2	0	1/12
1	0	0	1	1/12
1	0	2	0	1/12
1	0	2	1	1/12
1	1	0	1	1/12
1	1	1	0	1/12

$(X_1 | X_2 = 0) \sim \{0, 1\}$ with $\frac{(4/12)}{(7/12)}, \frac{(3/12)}{(7/12)}$

$(X_1 | X_3 = 0, X_4 = 1) \sim \{0, 1\}$ with $1/3, 2/3$

$X_3, X_4 | X_1 = 0$

$t_4 \backslash t_3$	0	1	2
0	1/7	1/7	2/7
1	1/7	2/7	0

$T_{X_4} = T_{X_2} = T_{X_1} = \{0, 1\}$ $\quad T_{X_3} = \{0, 1, 2\}$

Working

Conditioning and factors of the joint PMF

Suppose $X_1, X_2, X_3, X_4 \sim f_{X_1X_2X_3X_4}$ and $X_i \in T_{X_i}$.

$$f_{X_1 \cdots X_4}(t_1, \ldots, t_4) = P(X_1 = t_1 \text{ and } X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4)$$

Conditioning and factors of the joint PMF

Suppose $X_1, X_2, X_3, X_4 \sim f_{X_1 X_2 X_3 X_4}$ and $X_i \in T_{X_i}$.

$$\begin{aligned} f_{X_1 \cdots X_4}(t_1, \ldots, t_4) &= P(X_1 = t_1 \text{ and } X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4) \\ &= P(X_1 = t_1 \text{ and } (X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4)) \end{aligned}$$

Conditioning and factors of the joint PMF

Suppose $X_1, X_2, X_3, X_4 \sim f_{X_1X_2X_3X_4}$ and $X_i \in T_{X_i}$.

$$
\begin{aligned}
f_{X_1\cdots X_4}(t_1, \ldots, t_4) &= P(X_1 = t_1 \text{ and } X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4) \\
&= P(X_1 = t_1 \text{ and } (X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4)) \\
&= P(X_1 = t_1 | (X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4)) \\
&\qquad P(X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4)
\end{aligned}
$$

Conditioning and factors of the joint PMF

Suppose $X_1, X_2, X_3, X_4 \sim f_{X_1X_2X_3X_4}$ and $X_i \in T_{X_i}$.

$$
\begin{aligned}
f_{X_1\cdots X_4}(t_1,\ldots,t_4) &= P(X_1 = t_1 \text{ and } X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4) \\
&= P(X_1 = t_1 \text{ and } (X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4)) \\
&= P(X_1 = t_1 | (X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4)) \\
&\quad\ P(X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4) \\
&= P(X_1 = t_1 | X_2 = t_2, X_3 = t_3, X_4 = t_4)) \\
&\quad\ P(X_2 = t_2 | X_3 = t_3, X_4 = t_4) P(X_3 = t_3, X_4 = t_4)
\end{aligned}
$$

Conditioning and factors of the joint PMF

Suppose $X_1, X_2, X_3, X_4 \sim f_{X_1X_2X_3X_4}$ and $X_i \in T_{X_i}$.

$$
\begin{aligned}
f_{X_1\cdots X_4}(t_1, \ldots, t_4) &= P(X_1 = t_1 \text{ and } X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4) \\
&= P(X_1 = t_1 \text{ and } (X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4)) \\
&= P(X_1 = t_1 | (X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4)) \\
&\quad\quad P(X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4) \\
&= P(X_1 = t_1 | X_2 = t_2, X_3 = t_3, X_4 = t_4)) \\
&\quad\quad P(X_2 = t_2 | X_3 = t_3, X_4 = t_4) P(X_3 = t_3, X_4 = t_4) \\
&= P(X_1 = t_1 | X_2 = t_2, X_3 = t_3, X_4 = t_4)) \\
&\quad\quad P(X_2 = t_2 | X_3 = t_3, X_4 = t_4) \\
&\quad\quad\quad\quad P(X_3 = t_3 | X_4 = t_4) P(X_4 = t_4)
\end{aligned}
$$

Conditioning and factors of the joint PMF

Suppose $X_1, X_2, X_3, X_4 \sim f_{X_1X_2X_3X_4}$ and $X_i \in T_{X_i}$.

$$
\begin{aligned}
f_{X_1\cdots X_4}(t_1,\ldots,t_4) &= P(X_1 = t_1 \text{ and } X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4) \\
&= P(X_1 = t_1 \text{ and } (X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4)) \\
&= P(X_1 = t_1 | (X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4)) \\
&\qquad P(X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4) \\
&= P(X_1 = t_1 | X_2 = t_2, X_3 = t_3, X_4 = t_4)) \\
&\qquad P(X_2 = t_2 | X_3 = t_3, X_4 = t_4) P(X_3 = t_3, X_4 = t_4) \\
&= P(X_1 = t_1 | X_2 = t_2, X_3 = t_3, X_4 = t_4)) \\
&\qquad P(X_2 = t_2 | X_3 = t_3, X_4 = t_4) \\
&\qquad\qquad P(X_3 = t_3 | X_4 = t_4) P(X_4 = t_4) \\
&= f_{X_1|X_2=t_2,X_3=t_3,X_4=t_4}(t_1) f_{X_2|X_3=t_3,X_4=t_4}(t_2) \\
&\qquad\qquad f_{X_3|X_4=t_4}(t_3) f_{X_4}(t_4)
\end{aligned}
$$

Example: $X_1, X_2, X_3 \sim f_{X_1 X_2 X_3}$

t_1	t_2	t_3	$f_{X_1X_2X_3}(t_1, t_2, t_3)$
0	0	0	1/9
0	0	1	1/9
0	0	2	1/9
0	1	1	1/9
0	1	2	1/9
1	0	0	1/9
1	0	2	1/9
1	1	0	1/9
1	1	1	1/9

$$\underbrace{f_{X_1X_2X_3}(0,0,0)}_{1/9} = \underbrace{f_{X_3}(0)}_{3/9} \cdot \underbrace{f_{X_2|X_3=0}(0)}_{2/3} \cdot \underbrace{f_{X_1|X_2=0,X_3=0}(0)}_{1/2}$$

Working

Factoring can be done in any sequence

Suppose $X_1, X_2, X_3, X_4 \sim f_{X_1 X_2 X_3 X_4}$ and $X_i \in T_{X_i}$.

$$f_{X_1 \cdots X_4}(t_1, \ldots, t_4) = P(X_1 = t_1 \text{ and } X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4)$$

Factoring can be done in any sequence

Suppose $X_1, X_2, X_3, X_4 \sim f_{X_1X_2X_3X_4}$ and $X_i \in T_{X_i}$.

$$\begin{aligned} f_{X_1\cdots X_4}(t_1, \ldots, t_4) &= P(X_1 = t_1 \text{ and } X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4) \\ &= P(X_4 = t_4 \text{ and } X_3 = t_3 \text{ and } X_2 = t_2 \text{ and } X_1 = t_1) \end{aligned}$$

Factoring can be done in any sequence

Suppose $X_1, X_2, X_3, X_4 \sim f_{X_1 X_2 X_3 X_4}$ and $X_i \in T_{X_i}$.

$$
\begin{aligned}
f_{X_1 \cdots X_4}(t_1, \ldots, t_4) &= P(X_1 = t_1 \text{ and } X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4) \\
&= P(X_4 = t_4 \text{ and } X_3 = t_3 \text{ and } X_2 = t_2 \text{ and } X_1 = t_1) \\
&= f_{X_4 | X_3 = t_3, X_2 = t_2, X_1 = t_1}(t_4) f_{X_3 | X_2 = t_2, X_1 = t_1}(t_3) \\
&\qquad f_{X_2 | X_1 = t_1}(t_2) f_{X_1}(t_1)
\end{aligned}
$$

Factoring can be done in any sequence

Suppose $X_1, X_2, X_3, X_4 \sim f_{X_1 X_2 X_3 X_4}$ and $X_i \in T_{X_i}$.

$$
\begin{aligned}
f_{X_1 \cdots X_4}(t_1, \ldots, t_4) &= P(X_1 = t_1 \text{ and } X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4) \\
&= P(X_4 = t_4 \text{ and } X_3 = t_3 \text{ and } X_2 = t_2 \text{ and } X_1 = t_1) \\
&= f_{X_4 | X_3 = t_3, X_2 = t_2, X_1 = t_1}(t_4) f_{X_3 | X_2 = t_2, X_1 = t_1}(t_3) \\
&\qquad f_{X_2 | X_1 = t_1}(t_2) f_{X_1}(t_1)
\end{aligned}
$$

$$
f_{X_1 \cdots X_4}(t_1, \ldots, t_4) = P(X_3 = t_3 \text{ and } X_2 = t_2 \text{ and } X_1 = t_1 \text{ and } X_4 = t_4)
$$

Factoring can be done in any sequence

Suppose $X_1, X_2, X_3, X_4 \sim f_{X_1X_2X_3X_4}$ and $X_i \in T_{X_i}$.

$$\begin{aligned} f_{X_1\cdots X_4}(t_1,\ldots,t_4) &= P(X_1 = t_1 \text{ and } X_2 = t_2 \text{ and } X_3 = t_3 \text{ and } X_4 = t_4) \\ &= P(X_4 = t_4 \text{ and } X_3 = t_3 \text{ and } X_2 = t_2 \text{ and } X_1 = t_1) \\ &= f_{X_4|X_3=t_3,X_2=t_2,X_1=t_1}(t_4) f_{X_3|X_2=t_2,X_1=t_1}(t_3) \\ &\qquad f_{X_2|X_1=t_1}(t_2) f_{X_1}(t_1) \end{aligned}$$

$$\begin{aligned} f_{X_1\cdots X_4}(t_1,\ldots,t_4) &= P(X_3 = t_3 \text{ and } X_2 = t_2 \text{ and } X_1 = t_1 \text{ and } X_4 = t_4) \\ &= f_{X_3|X_2=t_2,X_1=t_1,X_4=t_4}(t_3) f_{X_2|X_1=t_1,X_4=t_4}(t_2) \\ &\qquad f_{X_1|X_4=t_4}(t_1) f_{X_4}(t_4) \end{aligned}$$

Example: $X_1, X_2, X_3 \sim f_{X_1X_2X_3}$

t_1	t_2	t_3	$f_{X_1X_2X_3}(t_1, t_2, t_3)$
0	0	0	1/9
0	0	1	1/9
0	0	2	1/9
0	1	1	1/9
0	1	2	1/9
1	0	0	1/9
1	0	2	1/9
1	1	0	1/9
1	1	1	1/9

$$\underbrace{f_{X_1X_2X_3}(0,0,0)}_{1/9} = \underbrace{f_{X_1}(0)}_{5/9} \cdot \underbrace{f_{X_2|X_1=0}(0)}_{3/5} \; \underbrace{f_{X_3|X_1=0,X_2=0}(0)}_{1/3}$$